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Molecular dynamics simulation of thiol-ene 'click' copolymerized and blended carboxymethyl cellulose and 4-arm polyethylene glycol hydrogel used as ciprofloxacin drug carrier

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ABSTRACT

While hydrogel-based drug carriers significantly enhance pharmaceutical efficacy, optimizing their nanoscale architecture remains a challenge for experimental observation alone. This study leverages all-atom molecular dynamics (MD) simulations to elucidate the distinct transport mechanisms of ciprofloxacin within thiol-ene 'click' carboxymethyl cellulose (CMC) and 4-arm polyethylene glycol (PEG) matrices. By contrasting copolymerized networks against physical blend composites, we identify a critical design trade-off: blend composites offer superior structural stability and stronger drug–polymer interactions, evidenced by lower root mean square deviations (RMSD) and radial distribution functions (RDF). Conversely, copolymer architectures demonstrate significantly higher swelling ratios and elevated mean square displacements (MSD), indicating enhanced water uptake and faster drug diffusion. These findings provide a molecular-level rationale for tailoring hydrogel formulations-selecting blends for sustained retention or copolymers for rapid release-thereby offering a predictive framework for the precise engineering of next-generation delivery systems.

1. Introduction

Drug delivery systems have emerged as a crucial field in pharmaceutical research, with sustainable hydrogel materials playing a crucial role due to their unique ability to control drug release rates and protect therapeutic agents (Ahmed, 2015; Hebeish and Sharaf, 2015; Muir and Burdick, 2021). Carboxymethyl cellulose (CMC)-based hydrogels, synthesized from renewable resources, have gained significant attention in drug delivery applications due to their biocompatibility, biodegradability, and non-toxicity (Kundu and Banerjee, 2020; Seera et al., 2021). Their excellent water retention capabilities and abundant hydroxyl groups, which enable chemical modifications, make them suitable carriers for both hydrophilic and hydrophobic drugs (Capanema et al., 2018; Hebeish and Sharaf, 2015; Kundu and Banerjee, 2020; Muir and Burdick, 2021; Pasqui et al., 2014). The formation of CMC hydrogels through thiol-norbornene 'click' chemistry presents a particularly promising approach, as this chemistry offers rapid reaction kinetics, high selectivity, and proceeds under mild physiological conditions without generating toxic byproducts (Capanema et al., 2018; Lowe, 2010). The thiol-ene reaction enables precise control over the crosslinking density and network structure, resulting in hydrogels with tunable mechanical and chemical properties (Hoyle and Bowman, 2010; Kazybayeva et al., 2022; Lowe, 2010). However, CMC hydrogels face certain limitations in drug delivery applications, including rapid degradation rates, inconsistent mechanical properties, and burst release of drugs, which can reduce their effectiveness as controlled release systems (Gupta et al., 2024; Nie et al., 2004). To address these challenges, the incorporation of 4-arm polyethylene glycol (4-arm PEG) as a crosslinked as well as blended with CMC has emerged as an effective strategy (Pettignano et al., 2019). The multi-arm structure of 4-arm PEG provides multiple crosslinking points, leading to enhanced mechanical stability and improved control over the degradation rate of the hydrogel network (Khan et al., 2022; Pettignano et al., 2019). The combination of CMC and 4-arm PEG creates a more robust and stable hydrogel system that maintains the beneficial properties of both components while mitigating their individual drawbacks. Furthermore, this hybrid hydrogel system contributes significantly to sustainability efforts in pharmaceutical development (Dadoo et al., 2017; Lee et al., 2016; Pettignano et al., 2019). The use of CMC, derived from renewable cellulose sources, reduces dependence on petroleum-based materials, while PEG, being biocompatible by the Food and Drug Administration (FDA), ensures minimal environmental impact (Capanema et al., 2018). The thiol-

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ene click chemistry employed in the synthesis process aligns with green chemistry principles, as it requires minimal energy input and generates negligible waste products (Kazybayeva et al., 2022; Lowe, 2010). The water-based preparation methods and biodegradable nature of these hydrogels further enhance their environmental sustainability credentials (Capanema et al., 2018). The resulting CMC-PEG hybrid hydrogel system represents a sustainable and efficient platform for drug delivery applications.

The CMC and PEG hydrogel systems can be tailored to accommodate various drug payloads (Ashour and Shafie, 2024; Capanema et al., 2018; Sanjanwala et al., 2022). Among them, the present study focuses on the encapsulation and controlled release of the antibiotic. Antibiotics are known to be ineffective in regenerative endodontic treatment because the drug degrades quickly (Kim et al., 2018; Sadek et al., 2019; Segura-Egea et al., 2017). Among them, ciprofloxacin is a widely used fluoroquinolone antibiotic renowned for its broad-spectrum antimicrobial activity and effectiveness against various bacterial infections. This drug has found applications in the treatment of respiratory, urinary tract, and gastrointestinal infections, as well as skin and soft tissue infections (Kaguelidou et al., 2011; Vance-Bryan et al., 1990). Ciprofloxacin also has disadvantages; its acidic nature can be harmful to stem cells (Kaguelidou et al., 2011). The CMC and PEG-based hydrogel are used to overcome the suboptimal drug concentration target site. The amphiphilic star polymers of four- and six-arm PEG with polycaprolactone (PCL), with different architectures, are synthesised and characterised, with ciprofloxacin loaded into the star polymer PCL-PEG hydrogels (Khodir et al., 2023; Lu et al., 2006; Singh et al., 2020). The presence of PEG in the outer block promotes blending, while the hydrophobic PCL tends to agglomerate and form non-homogeneous hydrogels. Both four-arm and six-arm star polymer hydrogels exhibit high loading efficiency (more than 90%). The drug release from the star polymer-based hydrogels follows the Korsmeyer–Peppas release profile, indicating sustained drug release (Khodir et al., 2023). The CMC-gelatin hydrogel crosslinked with Glutaraldehyde is synthesised and characterised by ciprofloxacin-loaded drugs. This gel shows a higher rate of drug release at a lower pH value (Sahar et al., 2023). The utilisation of CMC and PEG crosslinking in hydrogels serves as a potent drug delivery system, enhancing drug activity by facilitating controlled drug release at targeted sites (Lin et al., 2021; Sahar et al., 2023). Specifically, the thiol-norbornene crosslinked CMC-PEG hydrogel demonstrates notable capabilities in releasing Bovine Serum Albumin (BSA) with pH-dependent swelling behaviour at a pH of 4.0 (Lee et al., 2016). Moreover, the CMC-PEG hydrogel exhibits potential applications in wound dressing (Capanema et al., 2018). Despite the promising features of thiol-ene crosslinked CMC-PEG hydrogels observed in experimental literature, challenges persist. The high hydrophilicity of the network structure is intrinsic to such polymers and makes it difficult to determine the physicochemical properties accurately, and it often compromises reproducibility. The pivotal role of hydrogen bonding (H-bonding) in the application of hydrogels necessitates accurate mapping, a facet well-suited for theoretical investigations. Contrary to experimental limitations, molecular dynamics (MD) simulations offer a potent avenue to explore microstructures, discern material behaviours at the molecular level, and reveal essential physicochemical properties. In particular, MD simulations are central to elucidating the nature of key characteristics like structural orientations in solution media, prevailing nonbonded interactions like H-bonding, and transport characteristics of CMC and PEG hydrogel scaffolds.

MD simulations are essential for designing hydrogel-based drug delivery systems because they offer a thorough comprehension of atomic-level interactions and behaviours. A recent research suggests using multiscale computational pharmaceutics to assist formulation design and conduct a comprehensive analysis of formulation parameters of a common polymer-based drug delivery system (DDS), particularly hydrogels. The study recommends using MD simulations to predict the morphology and drug-releasing behaviour of hydrogels under various formulation factors at the formulation design stage, as well as to compute the solubility parameter (δ) and diffusion coefficient (D) for matrix screening of hydrogels at the pre-formulation stage. A number of characteristics, including the microstructure diagram, the radial distribution function (RDF), the radius of gyration (Rg), and the free diffusion volume, work together to describe the effect process. The usefulness of the approach is demonstrated by the high agreement between the simulated findings and in vitro release trials (Dai et al., 2022). MD simulations are conducted to study thiol-ene "click" hydrogels based on polyvinyl alcohol (PVA) and PEG. The MD analysis indicates that the diffusion coefficients for both the PVA and PEG hydrogels are consistent with the experimental diffusion coefficients (Das and Kundu, 2023, 2025b). This illustrates that the MD simulations help comprehend the processes of drug release, diffusion, and interactions in the hydrogel matrix, which in turn makes it easier to rationally design and optimise drug delivery systems based on hydrogel for improved therapeutic efficacy. In order to rationally develop and optimise hydrogel-based drug delivery systems for improved therapeutic effectiveness, MD simulations are a useful tool for clarifying their complicated behaviour.

In this study, four copolymer networks are developed by varying crosslinkers and their positions within the network. Additionally, four blending polymer models are formulated, altering these same parameters. Subsequently, the ciprofloxacin molecule is docked within the polymer network. Following the model preparation, the network is dissolved in water and a phosphate-buffered saline solution. Post-MD simulation, chronological trajectory analysis is conducted. Root mean square deviation (RMSD) is used to evaluate the structural stability. Understanding the interaction between polymers and solvents is crucial in hydrogel models, which involves computing H-bonding among diverse moieties and radial distribution functions. These computations elucidate the behaviour of polymer moieties within the solvent medium. Investigating transport properties entails computing the mean square displacements (MSD) of different polymer network moieties, enabling the determination of diffusion coefficients from the MSD curve.

2. Computational methodology

The monomeric structures of CMC and PEG are built with the Avogadro open source package (Hanwell et al., 2012). An in-house script is used to produce 5 copies of the CMC structure. The norbornene is functionalised in the third number of CMC to avoid the steric hindrance between the molecules. It is known as NCMC. Thereafter, thiol moieties are attached onto the norbornene group to form thiol-ene 'click' hydrogel. It is known as TCMC. The oligomeric CMC units are further functionalised with 3-mercaptopropionic acid to generate the mercaptopropionic acid functionalised CMC (SCMC). The representative structures are given in Fig. 1. Later, the NCMC, TCMC, and SCMC frameworks are replicated four times using an in-house program. The PEG monomer replicates twenty times using an in-house program, which is described in the literature (Das and Kundu, 2023; Deshmukh et al., 2024; Thombre et al., 2024). The end hydroxyl group of the PEG chain had been functionalised with norbornene for added benefits to the polymer framework (Lee et al., 2012; Lin and Lin, 2021; Shih and Lin, 2012). It is known as NPEG. In addition, the thiol moieties are incorporated in ene functionalisation to form a thiol-ene functionalised PEG hydrogel. It is known as TPEG. The PEG polymeric chain is functionalised with the 3-mercaptopropionic acid, which is known as the mercaptopropionic acid functionalisation of the PEG (SPEG). The four NPEG chains, TPEG chains, and SPEG chains are joined with a carbon atom in the Avogadro open-source package to create 4-arm-NPEG, 4-arm-TPEG, and 4-arm-SPEG chains. The experimental evidence suggests that the thiol group and ene group crosslink and generate the framework (Capanema et al., 2018; Lin and Lin, 2021; Shih and Lin, 2012). A couple of 4-arm NPEG chains and NCMC chains are crosslinked with a couple of TCMC chains, SCMC chains, 4-arm-TPEG chains, and 4-arm-SPEG chains. This cross-linking process helps to generate the four chains,

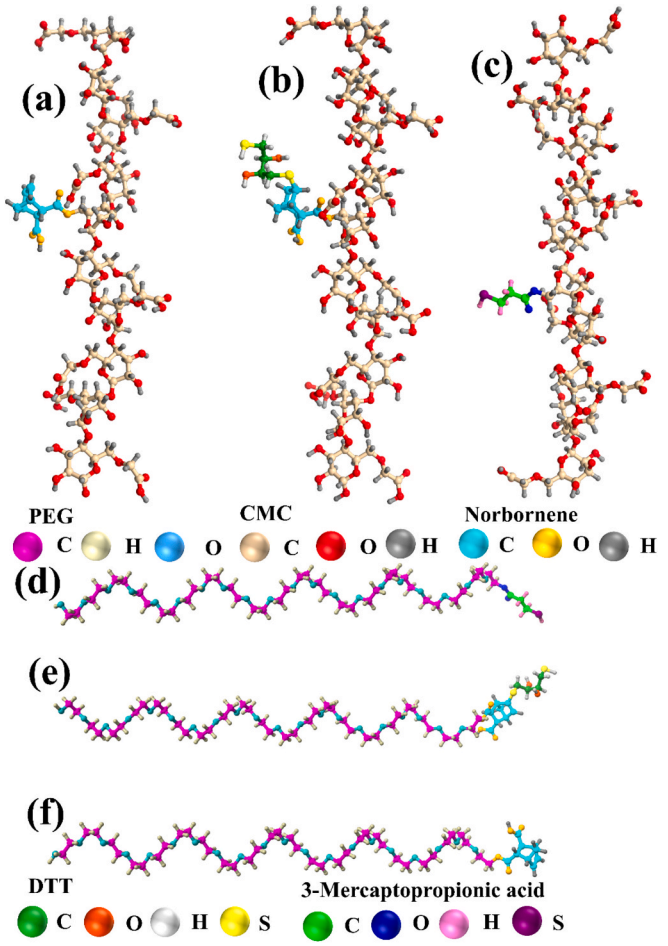


Fig. 1. Initial structure of blend polymer (a) NCMC (b) TCMC (c) SCMC (d) SPEG (e) TPEG (f) SPEG.

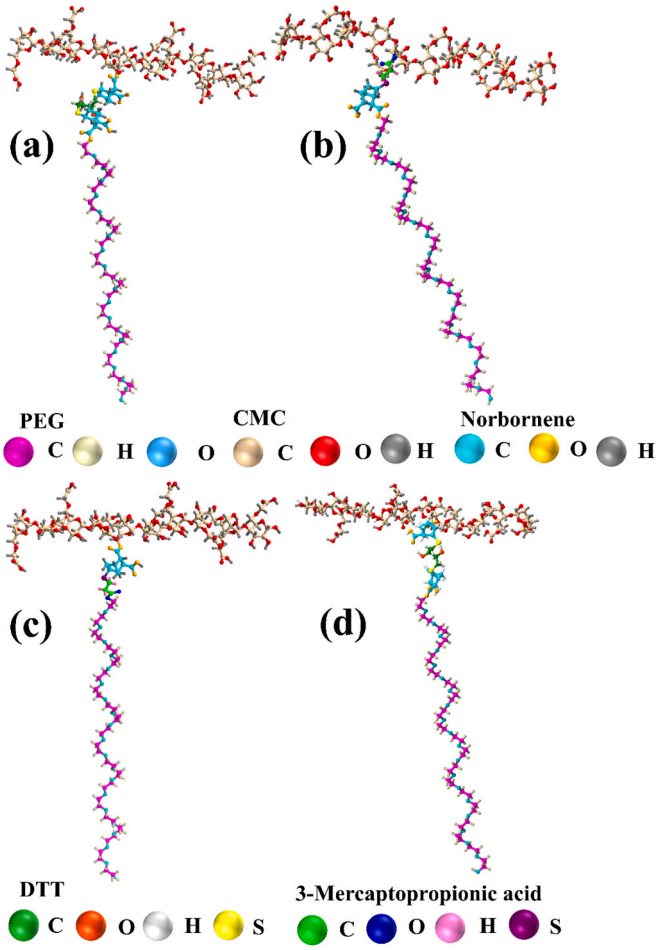


Fig. 2. Initial structure of copolymer composites (a) C-TCMC-NPEG (b) C-SCMC-NPEG (c) NCMC-SPEG (d) NCMC-TPEG.

which are 4-arm-NPEG copolymerise with TCMC (C-NPEG-TCMC), 4-arm-NPEG copolymerise with SCMC (C-NPEG-SCMC), NCMC copolymerise with 4-arm-TPEG (C-NCMC-TPEG), and NCMC copolymerise with 4-arm-SPEG (C-NCMC-SPEG). The copolymerise structures are shown in Fig. 2. Except for the copolymerisation structure, this study also incorporates similar blend systems, which are 4-arm-NPEG blend with TCMC (B-NPEG-TCMC), 4-arm-NPEG blend with SCMC (B-NPEG-SCMC), NCMC blend with 4-arm-TPEG (B-NCMC-TPEG), and NCMC blend with 4-arm-SPEG (B-NCMC-SPEG). The schematic diagram for building polymer frameworks is given in Fig. 3.

The four copolymer models and four blend polymer models are loaded with ciprofloxacin drugs using the Autodock Vina open-source package (Eberhardt et al., 2021; Gaillard, 2018; Trott and Olson, 2010). The work aims to study the diffusion of ciprofloxacin from the macromolecules to the solvent medium. To overcome the objective, the drug-loaded macromolecules are submerged in water medium and phosphate buffer saline (PBS) solutions. The PBS solution is prepared by mixing monosodium phosphate and bisodium phosphate in the aqueous solutions so that the system pH is maintained at 7.4. The PACKMOL open-source package has been used to generate the initial configuration of the systems (Martínez et al., 2009). The initial configurations of systems are given in Tables S1–S2 of Suporting Informations (SI). Bonded and non-bounded interactions have been evaluated by using the AMBER14SB force field (Maier et al., 2015). The Amber14sb force field parameter is primarily used to simulate protein–ligand interactions, but it is also applied in hydrogel simulations, which aligns with previous literature (Das and Kundu, 2023, 2025a, b). The functional form of the forcefield is founded on literature (Maier et al., 2015). To validate the forcefield, the density of CMC and PEG are calculated. The simulated density of the CMC is 1440 kg/m^3 and the experimental literature shows value is 1500 kg/m^3. The simulated density of CMC is lower than the experimental value because simulations use ideal polymer chains with uniform substitution and no bound water or counter-ions. Real CMC contains structural disorder, strong H-bonding, and residual moisture that allow tighter packing and increase density (Almeida and Silveira, 2023; Vyas et al., 2025). The simulated density of the PEG polymer is 1130 kg/m^3 and the experimental literature shows the value of 1100 kg/m^3, which is higher due to simulations often produce overly compact chain conformations with ideal branching and reduced free volume (Ho et al., 2024; Hoffmann et al., 2021). The dynamics of the system is investigated using an open source package, GROMACS. (Bauer et al., 2022).

The system's dynamics are acquired via energy minimisation, equilibration, and production steps. Employing a steepest descent algorithm with a 1000 KJ. mol^{-1}. nm^{-1} energy cut-off ensures MD simulation integrity. In the energy minimisation process, the maximum number of steps involved is half a million. During this process, the step size is taken for 0.001. The long-range electrostatic interactions are evaluated at a 1.2 nm cut-off distance. After that, the particle mesh Ewald algorithm is applied to evaluate the long-range electrostatic interactions (Cerutti et al., 2009; Ewald, 1921). The van der Waal (vdW) interactions are evaluated using a cut-off distance of 1.0 nm. Three-dimensional periodic boundary conditions are implemented. The leap-frog algorithm is used to integrate the equation of motion. The 2 fs time step is used for the isothermal-isobaric ensemble (NPT) and the canonical ensemble (NVT). The NVT and NPT ensemble is performed for 5 ns

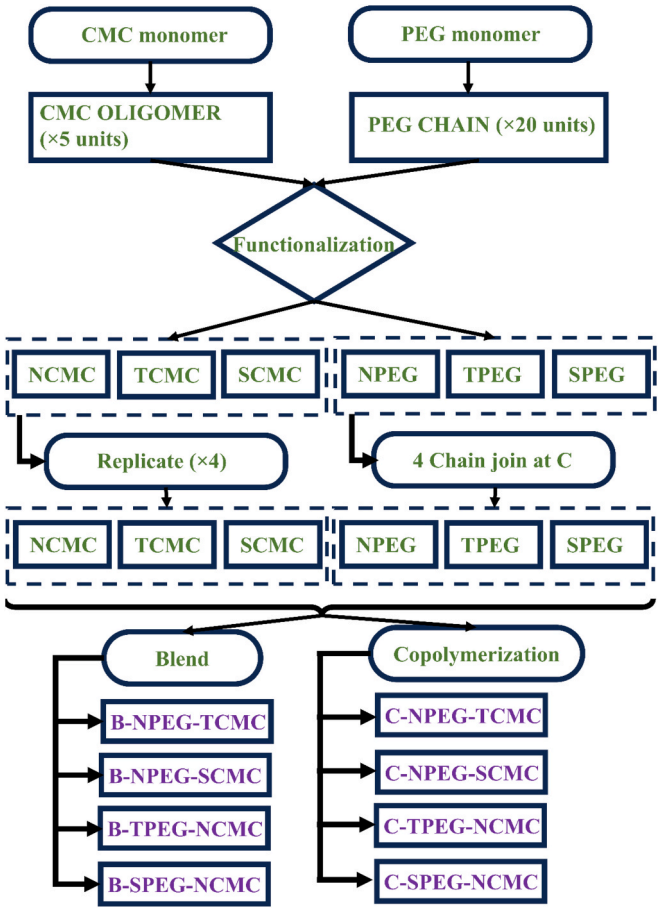


Fig. 3. Schematic representation of the polymer building framework.

with a system temperature of 310 K. The system temperature is kept constant using a velocity rescale thermostat (Bussi et al., 2007). During NPT, the system pressure is kept constant at 1.01325 bar using the Parinello-Raheman method (Parrinello and Rahman, 1981). The linear constraint solver algorithm was used for immobilising the movement of the hydrogen atoms (Hess et al., 1997). The production run is performed using the NPT ensemble. The production run is performed at 20 ns. During the production run, the trajectory is saved for 10 ps intervals. Trajectory visualisation is done using Chimera- X (Goddard et al., 2018; Pettersen et al., 2021) and VMD (Humphrey et al., 1996) open source packages and analysis is done using custom python scripts..

2.1. Trajectory analysis

2.1.1. Root mean square deviation

RMSD quantifies the structural deviation of a molecular system from a reference structure over time. To compute it, the average squared distances between matching atoms in the two structures under comparison are squared, and their square roots are taken. Mathematically, the RMSD formulae found in the literature (Das and Kundu, 2023).

2.1.2. Radial distribution function

The RDF is a key concept in MD simulations, as it gives information about the structural organisation and inter-molecular interactions in a system. It specifies the probability of finding particles at a particular distance from a fixed particle. The mathematical form of the RDF is mentioned in literature (Das and Kundu, 2023; Deshmukh et al., 2024).

The average number of neighbours i.e. numbers of particles or atoms within a predefined cut off distance from a central particle is known as the coordination number (*CN*). It provides information about the local environment and packing arrangements of the simulated system. The *CN* is then obtained by integrating the RDF to the first minimum. The mathematical formulation of the *CN* is also available in literature (Das and Kundu, 2023; Deshmukh et al., 2024).

2.1.3. H-bonding

H-bonding is one of the most important intermolecular interactions in MD simulations, particularly for simulations involving systems that consist of polar molecules or biomolecules. An H-bond is developed when one hydrogen atom covalently attached to an electronegative donor atom interacts with an electronegative acceptor with electrostatic attraction. In MD simulations, H-bonds have usually been recognized by geometrical criteria as described in literature (Arunan et al., 2011).

2.1.4. Mean square displacements

The MSD describes the distance travelled by particles or atoms in a MD simulation as a function of time. It is a useful metric for characterising the diffusive behaviour and mobility of the system. The MSD is calculated as the average squared displacement of particles from their initial positions over a given time interval. The mathematical form of MSD is found in the literature (Maginn et al., 2019). The MSD curves are characterised by the subdiffusive ($\beta < 1$), superdiffusive ($\beta > 1$) and diffusive ($\beta = 1$) behaviour. The mathematical formulae are given bellow

$$\beta = \frac{\log(MSD_2) - \log(MSD_1)}{\log(t_2) - \log(t_1)} \tag{1}$$

2.1.5. Swelling ratio

The swelling ratio is a major parameter in the MD simulations of hydrogels because it describes the ability to absorb and retain large quantities of water or other solvents, which in turn directly affects their structural and functional properties. It quantifies the degree of swelling or the amount of solvent uptake by the hydrogel or polymer system. The swelling ratio is computed by calculating the amount of freezing water and nonfreezing water present in the system. Non-freezing water is defined as the amount of water molecules directly H-bonded with the polymer scaffolds. The freezing water is defined as the amount of water that is attached to the non-freezing water by virtue of H-bonds (Jiang et al., 2017). As the number of non-freezing water molecules increases, osmotic pressure rises while elastic interactions decrease. As the amount of non-freezing water increases, the degree of physical crosslinking decreases, which reduces the mechanical strength of the matrix, which also impacts the elastic interactions. As the physical crosslinking increases, it reduces the availability of hydrophilic sites for water, thus decreasing osmotic pressure, which impacts the swelling ratio. Non-freezing water is associated with non-freezing H-bonds between the polymer matrix and solvents, which directly influences one of the computation swelling ratio. The mathematical form of the swelling ratio is found in the literature (Das and Kundu, 2023).

3. Result and discussion

3.1. Identification of drug binding sites in the polymer matrix

The auto dock score is given in Tables 1 and 2 for the copolymer and

Table 1
Docking score of copolymer systems.

System name	Affinity (Kcal/mol)	distance from best mode	
		rmsd l.b.	rmsd u.b.
C-NCMC-TPEG	−5.9	0.0	0.0
C-NCMC-SPEG	−6.2	0.0	0.0
C-TCMC-NPEG	−5.9	0.0	0.0
C-SCMC-NPEG	−5.8	0.0	0.0

Table 2
Docking score of the blend polymer.

System name	Affinity (Kcal/mol)	distance from best mode	
		rmsd l.b.	rmsd u.b.
NCMC	−6.6	0.0	0.0
TCMC	−7.0	0.0	0.0
SCMC	−6.2	0.0	0.0
NPEG	−4.8	0.0	0.0
TPEG	−5.4	0.0	0.0
SPEG	−5.1	0.0	0.0

blend systems, respectively. The highest docking score is observed in the C-NCMC-SPEG system (−6.2 Kcal/mol) for the copolymer systems. This happens due to H-bonding between drug moieties and polymer matrix whereas the alternate copolymer system is free from such interactions with drug moieties. However, the other copolymer system does not form H-bonds with the drug moieties. The drug moieties interact with the polymer matrix using weak vdWs interactions. When changing the functionalisation from the CMC matrix to the PEG matrix, the docking score decreases in the C-NCMC-SPEG and C-SCMC-NPEG systems. When the norbornene moiety functionalises with CMC, the hydrophobicity of the polymer matrix increases. As a result, the drug easily goes into the inside of the polymer matrix and forms H-bonds, as shown in Fig. 4. The blend system is made by mixing two polymer matrices. So the docking score for ciprofloxacin is performed for each polymer matrix. The TCMC system has shown the highest docking score (−7.0 Kcal/mol), which is due to the H-bonding between the polymer matrix and drug moieties, as shown in Fig. 5. However, the PEG functionalised polymer matrix shows a lower docking score compared to the CMC functionalised polymer matrix. PEG's flexible, non-ionic nature often leads to reduced interactions with drug molecules, potentially limiting strong H-bonding or electrostatic interactions (Parray et al., 2020). In contrast, CMC contains carboxyl groups, which can form H-bonds or ionic interactions with ciprofloxacin, enhancing its binding affinity. However, the single polymer matrix shows a higher docking score compared to the copolymer composites. The copolymer matrix is formed by crosslinking two polymer matrices. During the crosslinking process, the active site of the polymer matrix is blocked. As a result, the binding affinity of the copolymer systems is reduced.

3.2. Stability of the drug molecules in the polymer matrix

The RMSD of polymer-chain-drug molecules to polymer-chain-drug molecules are shown in Fig. 6 and Figs. S1–S2 of SI. Fig. 6(a), Fig. S1 (a) and Fig. S2(a) show the RMSD between the blend composites in the PBS solution. The RMSD increases until 9.0 ns, after which it deviates from the mean value for the B_NCMC_TPEG, B_TCMC_NPEG, and B_SCMC_NPEG systems. The RMSD value is 1.0 nm for the three blend systems in the PBS medium. However, the RMSD of the B_NCMC_SPEG system increases by 18 ns, where the RMSD value is 3.0 nm. After that, the RMSD value decreased to 2.3 nm for the 24 ns. After that, the RMSD value deviates from the mean value. This suggests the B_NCMC_SPEG system is more unstable than the other three systems in the PBS medium. This further suggests the B_NCMC_SPEG blend system can easily release the ciprofloxacin drug molecules in the PBS medium compared to the other blend systems in the PBS medium. Fig. 6(b) and S1–S2(b) suggest the RMSD between the ciprofloxacin-blend system in the aqueous medium. The RMSD of the B_TCMC_NPEG system is lower compared to the other systems. The RMSD of the B_TCMC_NPEG system is initially increased for two ns. After that, it deviates from the mean value till the end of the simulation. The RMSD value for this system is 0.23 nm at the end of the simulation, which suggests the drug molecules are strongly attached to this system. This suggests the B_TCMC_NPEG system can hold the ciprofloxacin for longer time periods. A further detailed

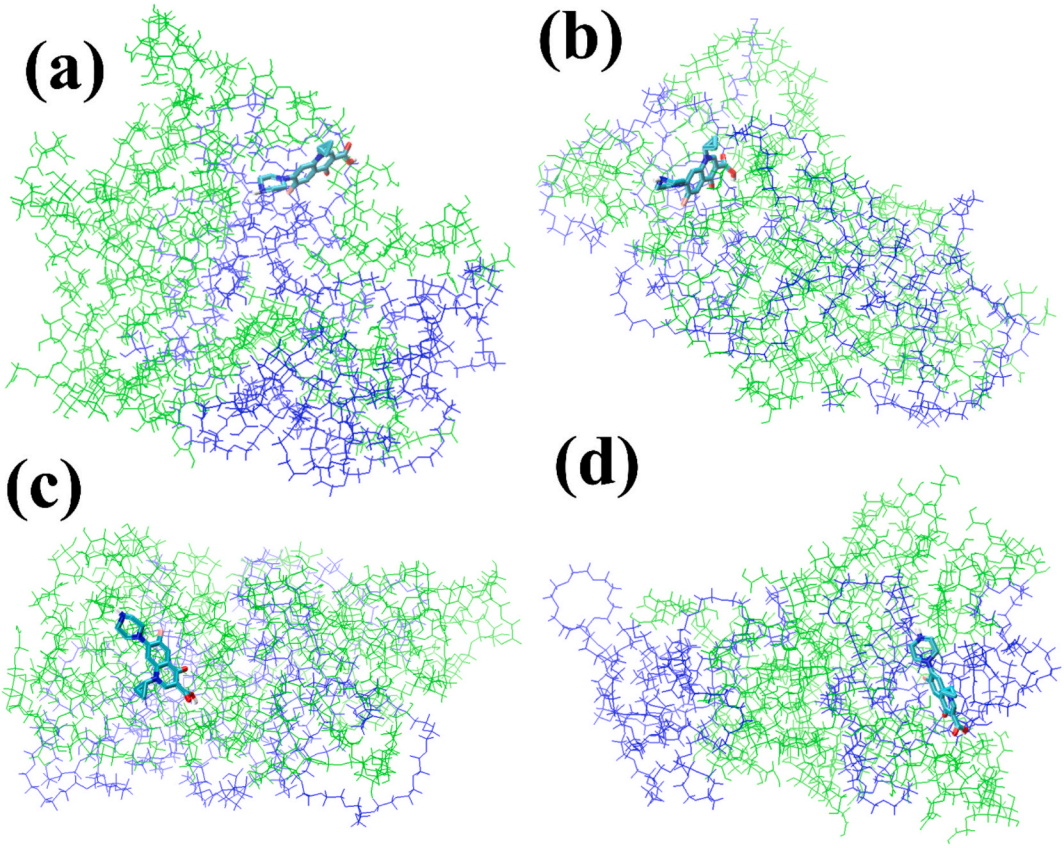


Fig. 4. Molecular docking of ciprofloxacin in the different copolymers (a) C-NCMC-TPEG (b) C-NCMC-SPEG (c) C-TCMC-NPEG (d) C-SCMC-NPEG.

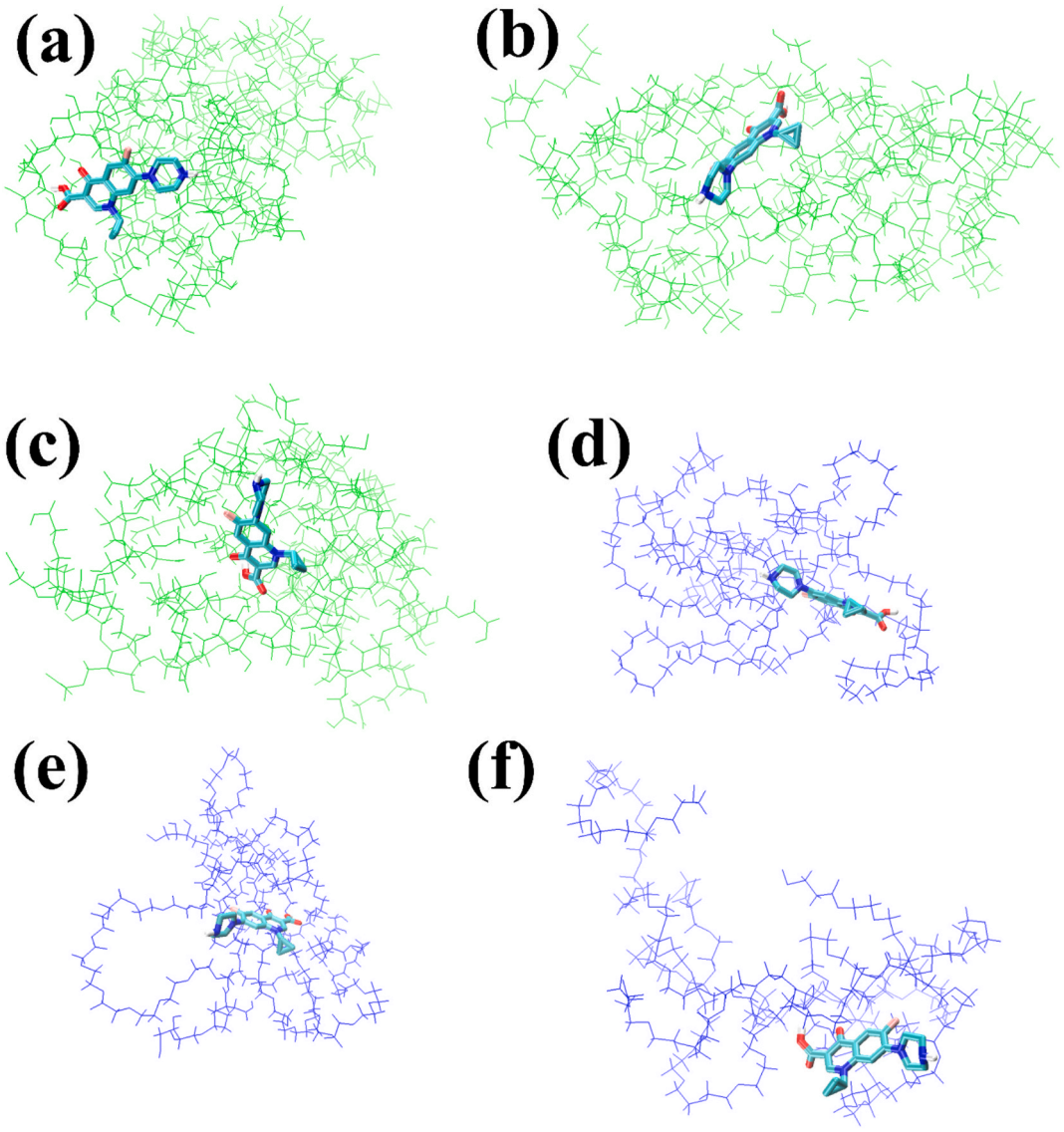


Fig. 5. Molecular docking of ciprofloxacin in the different homopolymers (a) NCMC (b) TCMC (c) SCMC (d) NPEG (e) TPEG (f) SPEG.

description of this system is in the upcoming sections. However, the other three systems RMSD fluctuates throughout the dynamics, which suggests the instability of the systems. The RMSD of the B_SCMC_NPEG system is showing a higher RMSD than that of the other systems. The RMSD of the B_SCMC_NPEG system shows non-linear increments of RMSD with respect to time. It suggests the ciprofloxacin molecules are more strongly interacting with the solvent molecules than the blend matrix. By changing the crosslinker position from the CMC polymer matrix to the PEG polymer matrix, the RMSD results significantly changed, which is due to the hydrophilicity and hydrophobicity of the molecules. Fig. 6(c) is the RMSD of the ciprofloxacin-copolymer matrix in the PBS system. The RMSD of ciprofloxacin in the C-NCMC-TPEG and C-NCMC-SPEG shows similar trends with respect to time, and the values are 2.2 nm and 2.5 nm, respectively. The RMSD of ciprofloxacin in both systems initially increases in nature; then, it deviates from its mean value. It suggests the non-bonded interactions between the drug molecules and the copolymer matrix. The RMSD of ciprofloxacin in the C-TCMC-NPEG and C-SCMC-NPEG shows chronological, linear increments till 24 ns and 20 ns, respectively. It suggests that ciprofloxacin interacts more with the PBS system than the copolymer matrix. After that time period, the RMSD becomes linear for both systems. It suggests non-bonded interactions between the drug molecules and the copolymer matrix. The RMSD of ciprofloxacin in the C-TCMC-NPEG system has a higher value compared to other matrix, which suggest the drug molecules are moving from their initial positions, as explained in the further sections. Comparing Fig. 6(a) and (c), the RMSD values are higher in the copolymer system. The Drug molecules are bound by the non-bonded interactions; as a result, the RMSD is higher in the copolymer system, which is validated by the docking results. This suggests the drug moieties are more stable in the blend system. This further suggests that the drug release from the copolymer system is easier compared to the blend systems. Fig. 6(d) is the RMSD of ciprofloxacin in the copolymer matrix in the aqueous medium. The RMSD of ciprofloxacin in the C_TCMC_SPEG and C-TCMC-NPEG shows similar behaviour throughout the simulations. In the beginning, the RMSD rises for the first seven nanoseconds. After that, the RMSD curve reaches a level until the end of the simulation. The first increase in RMSD is a conformational change in the aqueous medium due to the presence of hydrogenous moieties in the system. The plateau phase suggests the structural status of ciprofloxacin in the aqueous environment. The RMSD of ciprofloxacin in the C-SCMC-NPEG system shows linear increments of RMSD throughout the simulation. It suggests the drug molecules are moving from binding sites to the solvent medium gradually. The RMSD in the C-NCMC-TPEG system initially shows the deviation from the mean value till 17 ns. It suggests

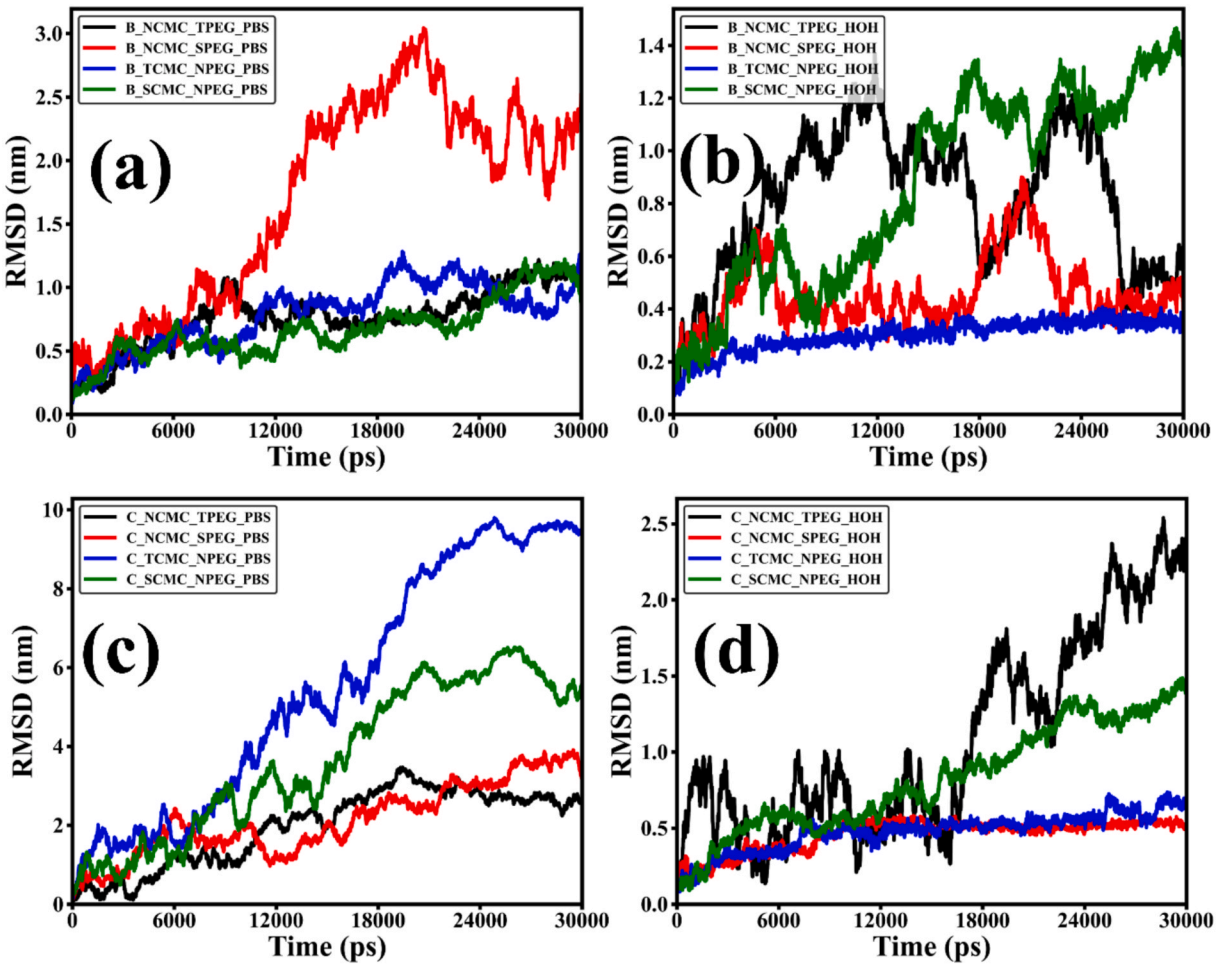


Fig. 6. RMSD of polymer framework to drug molecules (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous.

the drug molecules are interacting with the other polymer moieties from the binding sites. Later, the RMSD shows linear increments to the end of the simulation which implies that the drug molecules lose contact with the binding sites. The change of solvents from water to PBS significantly changes the RMSD value. It suggests the drug molecules have more mobility from their initial binding sites in the PBS medium. This is due to the interaction of the salts present in the PBS between the drug molecules. The drug molecule in the copolymer shows more RMSD than the blend system. It suggests drug molecules are weakly non-bonded and interact with the copolymer, which is discussed in further sections. Figs. S3–S5 of SI shows the RMSD of the polymer matrix to the polymer matrix. It is observed that the drug-polymer matrix RMSD is higher compared to the polymer matrix. This suggests the instability of the system. This suggests the drug molecules can easily be removed from hydrophobic cores.

3.3. Stability of the hydrogel matrix

Figs. 7, S7–S8 shows the average number of H-bonds between the polymer to the solvents, polymer1 to solvents and polymer2 to solvents, respectively. Fig. 7(a) shows the H-bonds between the blend composites and PBS mediums. The TCMC_NPEG system forms a higher number of H-bonds due to the presence of dithiothreitol moieties. These moieties enhance the solubility of CMC in aqueous media, increasing the overall H-bonding capacity. Figs. 6(a), S7(a) and S8(a) show the stable H-bond formation, which signifies the formation of gel networks. However, there is a slightly negative slope that indicates increased stability of the hydrogel network in PBS media(Zhang et al., 2025). As the number of H-bonds decreases, it enhances the number of H-bonds between the polymer matrix, which enhances the physical crosslinking process and also impacts the self-healing property (Yu et al., 2022; Zhang et al., 2025). It also signifies the reduction of drug loading capacity with respect to time and slows down the degradation process (Yu et al., 2022; Zhang et al., 2025). Fig. 7(b) shows the average number of H-bonds between the blend polymer composite and the water mediums. It shows a negative slope with dynamics. This is due to the thiol-ene crosslinked network, which might expose hydrophobic segments, reducing the availability of H-bonding sites. This suggests the reduction of the water-holding capacity of polymer scaffolds. As a result, this can reduce the mechanical strength and stability of the hydrogel. To further understand the role of CMC functionalised and 4-arm-PEG functionalised polymer matrix interaction with the water, it can be visualised in Figs. S7(b) and S8(b), respectively. The CMC functionalised polymer matrix is showing a negative slope, which signifies the enhanced stability of hydrogel networks. Comparing Fig. 6(a) and (b) shows that the blend composites are forming more H-bonds in PBS media compared to aqueous media. This phenomenon has been attributed to the presence of interactions between ions existing in the PBS medial, that contribute to the increase in the stability of the hydrogel matrix. Fig. 6(c) shows the time evolution of 368 hydrogen bonds with PBS in the matrix of the copolymer, which shows a stable hydrogen bond network and thus formation of a strong gel. The TCMC_NPEG system shows an increased number of hydrogen bonds due to the dithiothreitol moieties. Comparing Fig. 6(a) and (c) shows that the blend system contains more H-bonds than the copolymer

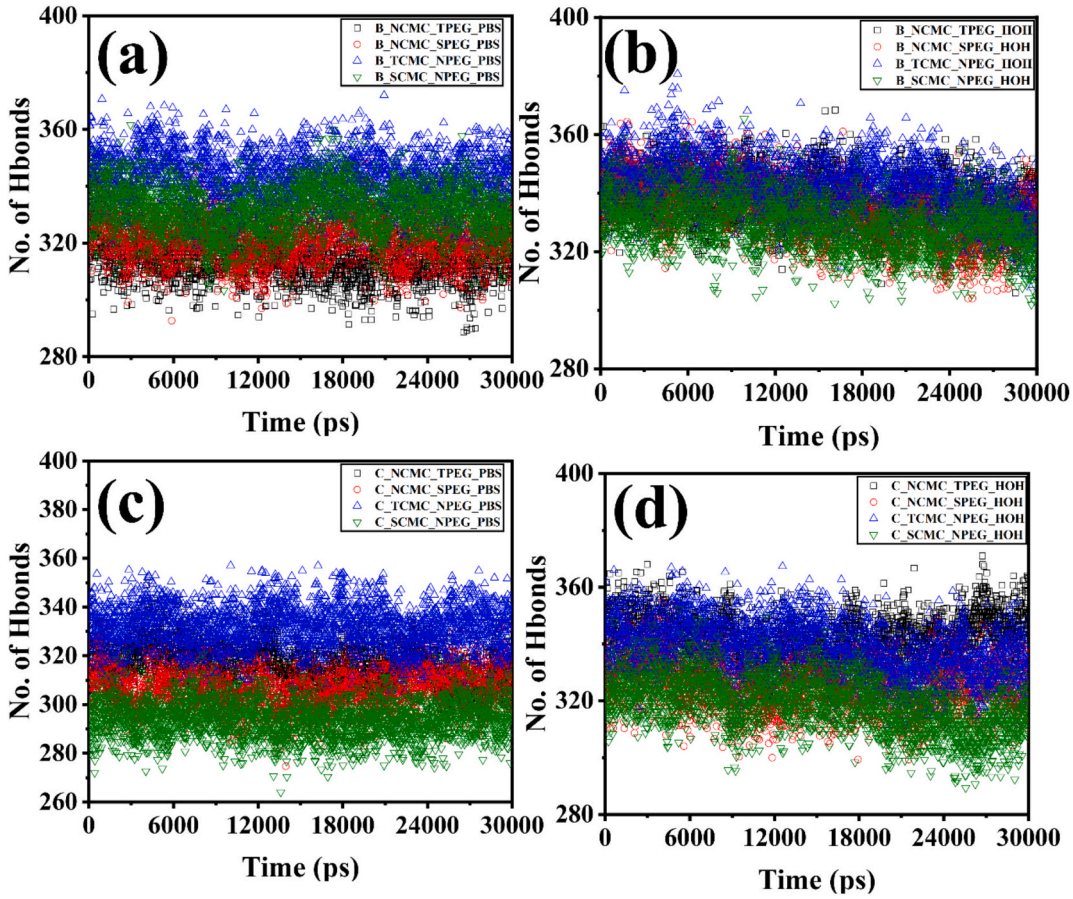


Fig. 7. H-bond of polymer framework to the solvent (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous.

system, due to copolymerisation process blocking the active H-bonding procreation. Fig. 6(d) shows the average number of H-bonds that are formed between the copolymer composite and the aqueous medium. The negative slope of the curve suggests a reduction in polymer stability in the aqueous medium, likely due to the thiol-ene crosslinked network exposing hydrophobic segments and reducing H-bonding site availability.

Fig. 8 highlights the interchain and intrachain H-bonds within the polymer matrix. Fig. 8(a–d) depict the average number of H-bonds in blend composites and copolymer composites in PBS and aqueous media, with data summarized in Table S3. The TCMC_NPEG system consistently shows higher H-bond numbers due to the dithiothreitol moiety, which offers additional donor and acceptor sites compared to mercapturic acids. Among the two dithiothreitol chains, one forms fewer H-bonds as its active sites are blocked by solvent interactions. Fig. 7(a) and (b) reveal that aqueous media slightly increase H-bonds compared to PBS, owing to enhanced polymer–solvent interactions. Comparing Fig. 8(a) and 8(c) demonstrates that the blend systems form more H-bonds between polymers than copolymer systems, a trend also seen in Fig. 7(b) and (d). This difference arises as copolymer composites prioritise interactions with solvents, reducing polymer–polymer H-bonds. Fig. 8(a), (b), and (d) show increasing H-bonds over time, enhancing polymer matrix stability but potentially reducing hydrogel mechanical stability. Fig. S9 further explores H-bonds between polymer matrix 1 (CMC-functionalized) and polymer matrix 2 (PEG-functionalized), with a range of 10–40H-bonds, as detailed in Table S1. Copolymer composites exhibit more H-bonds between these matrices than blends due to crosslinking, which brings polymer chains closer and exposes additional bonding sites. This structural arrangement facilitates higher H-bond formation in copolymer systems with solutions, supporting stability trends observed across the matrices.

Fig. S10 illustrates the number of H-bonds between the drug and solvents. In the blend composites, the drug forms an average of 3.25H-bonds with water and 4.0 with PBS, indicating stronger interactions in PBS and faster drug release compared to water. Conversely, in copolymer composites, the average H-bonds with both water and PBS are 3.0, suggesting that changing the medium does not impact drug release. Drug molecules have a greater number of hydrogen bonds with the solvent than with the polymer matrix. This indicates a stronger affinity of the drug towards the solvent environment, suggesting enhanced solubility. However, this may also imply reduced interaction and compatibility with the polymer carrier over time. This observation is further supported and elaborated upon in the MSD results section. Fig. S11 shows H-bonds between the polymer and the drug matrix. The blend composite forms a higher number of H- bonds than the copolymer matrix primarily due to its structural flexibility and molecular mobility. In the blend system, small and discrete polymer chains can rearrange more freely during the MD simulation, allowing functional groups to optimally orient toward the drug molecules. This enhanced conformational freedom increases the accessibility of hydrogen bond donors and acceptors at the polymer-drug interface. In contrast, the copolymer matrix has a more constrained architecture, limiting such rearrangements. The docking results support this behavior by showing stronger and more frequent polymer-drug interactions in the blend matrix. Supporting Data in Figs. S12 and S13 reveal that CMC chains form more H-bonds with drugs than 4-arm PEG chains, further highlighting the influence of polymer type on drug interactions.

These findings provide critical engineering insights for designing efficient drug delivery systems. The higher H-bonding capacity of TCMC_NPEG in PBS indicates improved hydrogel stability and

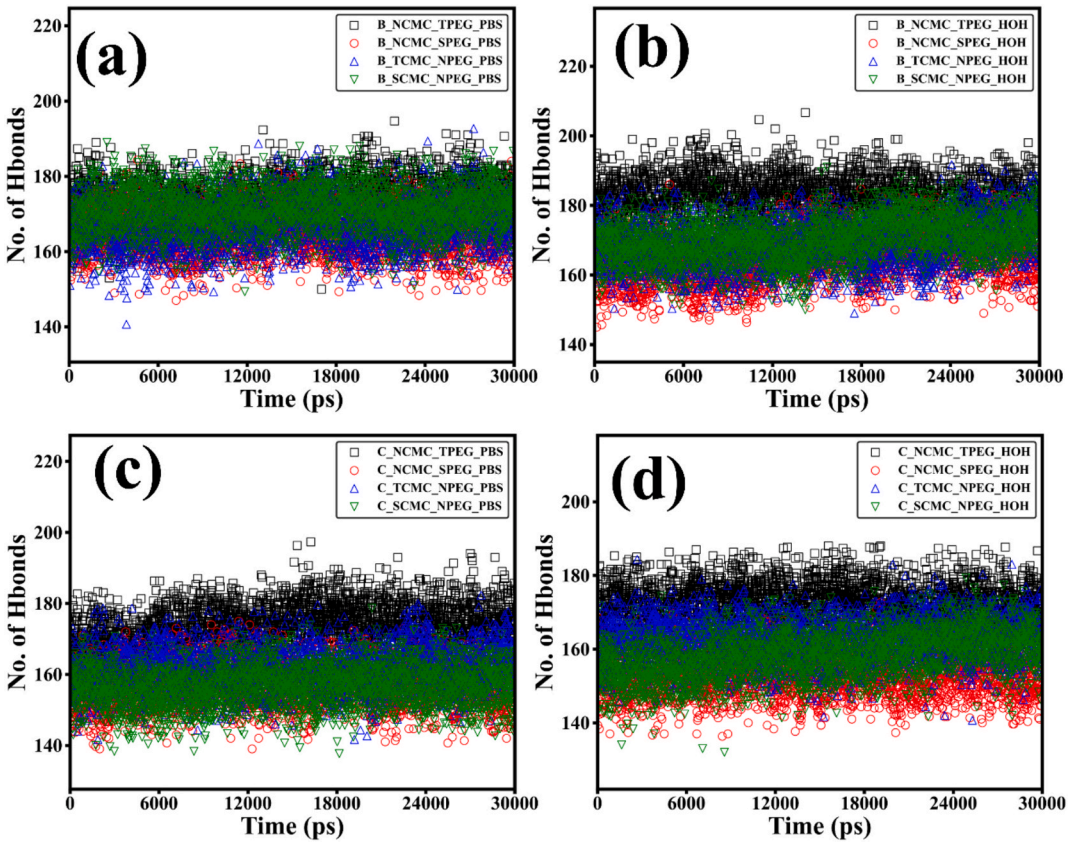


Fig. 8. H-bond of polymer framework to the polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous.

controlled degradation, making it ideal for sustained drug release. However, excessive H-bonding may limit drug loading and mechanical flexibility. The reduced H-bonds in aqueous media suggest decreased water retention, influencing swelling behavior and release kinetics. Blend systems, with their enhanced polymer–drug H-bonding and conformational flexibility, enable better drug encapsulation and responsiveness, while copolymer systems offer structural stability. Balancing H-bond density, crosslinking degree, and solvent interactions is thus key to optimizing drug release and hydrogel durability.

3.4. *Swelling ratio of the hydrogel networks*

Fig. 9 shows the swelling ratio of the hydrogel models in the last frame. Fig. 9(a) and (b) show the swelling ratio of the blend hydrogel in the PBS and aqueous mediums. The B-NCMC-SPEG system shows a higher swelling ratio in the PBS system with a mean value of 136.84 and a standard deviation of 4.97. The higher swelling ratio suggests a higher drug loading capacity compared to the other hydrogel models, which is further discussed on the next section in the results. This is due to the absorption of the freezing water more than the non-freezing water. The B-NCMC-SPEG system is made of norbornene functionalised CMC and mercaptronic acids functionalised 4-arm PEG chain. The norbornene functionalises the CMC chain containing a carboxylic acid group, which can absorb a higher amount of freezing and non-frezzing water. The mercaptronic acids function by forming a 4-arm PEG chain with h-bond acceptors, which suggests a higher number of absorbing non-freezing and freezing waters. The B-TCMC-NPEG system shows a higher swelling ratio in the aqueous medium, as shown in Fig. 9(b). The mean swelling ratio value is 145.87 with a standard deviation of 2.7. The H-bonds results suggest the B-TCMC-NPEG system has a higher number of H-bonds, which suggests the absorption of the non-freezing water. As the system has a higher amount of non-freezing water, it can also absorb a higher amount of freezing water. Comparing Fig. 9(a–b) suggests the blend composites are forming a higher swelling ratio in water medium compared to the PBS mediums. However, the value is slightly higher in the aqueous medium, which also suggests the H-bonding results. Fig. 9 (c) and (d) suggest the swelling ratio of the copolymer composites in PBS and aqueous mediums, respectively. The C-NCMC-TPEG and C-SCMC-NPEG systems show similar swelling ratio values in the PBS system. The average swelling ratio value is 354.70, with a standard deviation of 14.90 for the C-NCMC-TPEG system. The C-NCMC-TPEG system contains norbornene fuctionalises CMC and dithiothreitol functionalises with 4-arm-PEG chains. The norbornene functionalises CMC, which contains carboxylic acids and hydroxyl, which can help to absorb the freezing and non-freezing water. The TPEG chain contains the dithiothreitol moiety, which has a couple of hydroxyl groups that can help to enhance the absorption of freezing and non-freezing water. Fig. 9(a) and (c) shows that copolymer composites have a higher swelling ratio than blend composites. The copolymer composites absorb a higher amount of freezing water compared to blend composites. The blend composites are forming more H-bonds with the polymer as a result, which shows a lower swelling ratio. Fig. 9(d) shows the swelling ratio of the copolymer composites in the aqueous mediums. The C-NCMC-SPEG polymer chain shows a higher swelling ratio compared to the other chains in aqueous media. The swelling ratio of the C-NCMC-SPEG is 414.14 with a standard deviation of 26.72. These polymer composites are showing higher swelling ratios due to the norbornene and mercaptronic acid functionalisations in the polymer chains. The norbornene has carboxylic acids and unsaturated hydrocarbons, which can help to absorb the freezing water and non-freezing water. Mercaptonic acids contain thiol and carboxylic acids, which also help absorb water. Comparing Fig. 9(c) and (d), it shows that the copolymer in the aqueous medium shows a higher swelling ratio, which suggests the higher stability of the copolymer in the aqueous medium. The RMSD results also conclude similar

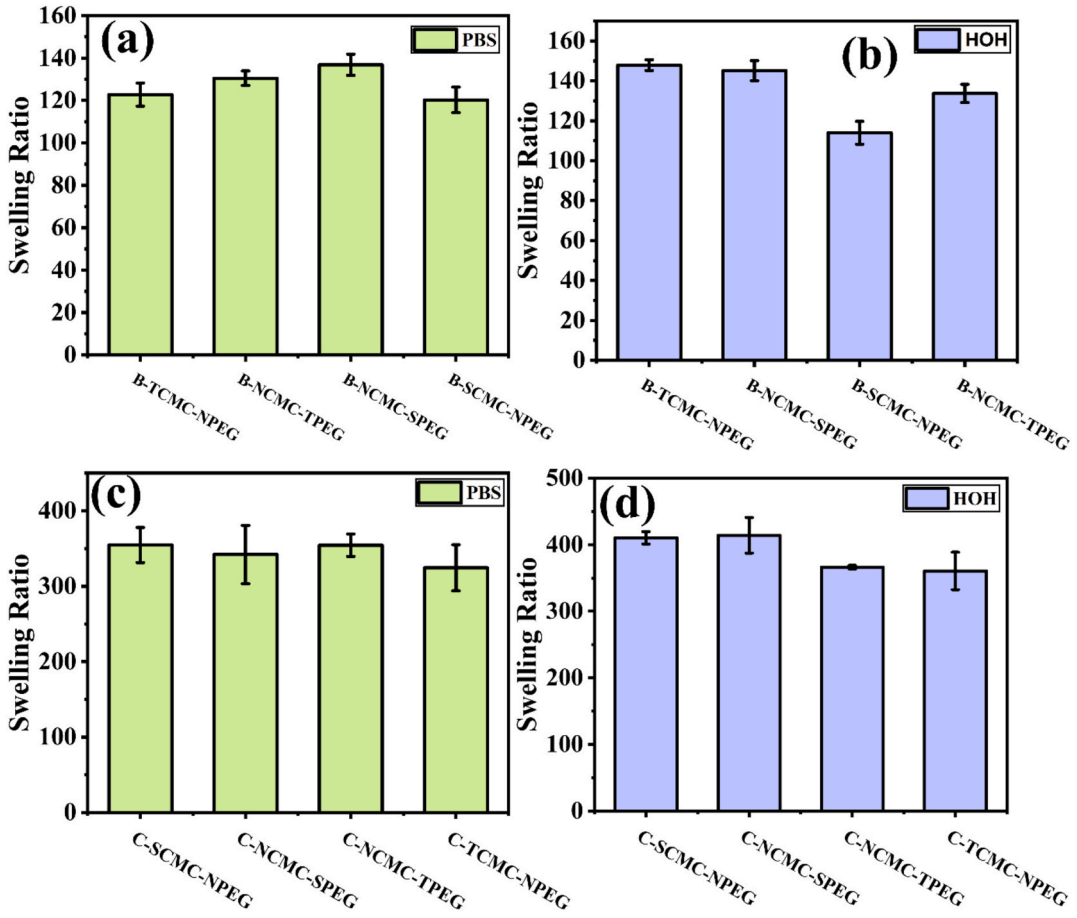


Fig. 9. Swelling ratio polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous.

behaviours. Comparing Fig. 9(b) and (d), the copolymer has a higher swelling ratio compared to the blend hydrogel models. Due to the crosslinking process, the copolymer makes bulky macromolecules. During the simulation, the copolymer structures themselves assemble so that the hydrophobic moieties are present in the core and the hydrophilic moieties are present on the surface. A hydrophilic moiety can help absorb freezing and non-freezing water.

These swelling behavior insights are vital for engineering advanced hydrogel-based drug delivery systems. The higher swelling ratios in copolymer systems, especially C-NCMC-SPEG, indicate superior water absorption and drug loading potential, essential for prolonged and controlled drug release. Functional groups like carboxylic acids, hydroxyls, and thiols enhance hydrophilicity and swelling, optimizing drug diffusion and bioavailability. Conversely, blend hydrogels, with lower swelling ratios, offer greater structural integrity and slower release rates. Thus, tuning functional group composition and crosslinking density enables precise control over swelling, stability, and release kinetics, ensuring tailored hydrogel performance for specific therapeutic applications and environmental conditions.

Interactions of the drug moiety with the solvent, hydrogel surface, and intra- and inter-network interactions within the hydrogel:

Fig. 10 shows the RDF between the oxygen of the hydroxyl group in ciprofloxacin and the water hydrogen. The solvation shells and *CNs* are shown in Table S4. The first solvation shell is due to the H-bonding regions. This suggests that the hydroxyl moiety strongly interacts with water, which can help to remove drug moieties from its binding sites. After the first solvation shell, the RDF shows an upward straight trend, which signifies a lack of well-defined solvation layers beyond the first shell. This could indicate that the system transitions into a bulk-like behaviour, where the spatial organisation of particles becomes more uniform and random. Fig. 10(a) shows the B_NCMC_TPEG_PBS system is showing a higher *CN* number compared to the other systems, which suggests a higher number of water molecules surrounded by this moiety. This also suggests the stronger interactions of the moiety compared to the other systems. Fig. 10(b) shows the B_SCMC_NPEG_HOH system is showing a higher *CN* number compared to the other systems, which suggests a higher number of water molecules surrounded by this moiety. This also suggests the stronger interactions of the moiety compared to the other systems. Comparing Fig. 10(a and b), the significant change of RDF values in the B_NCMC_TPEG and B_SCMC_NPEG changes in the PBS to aqueous mediums. This suggests the self-assembly of the structures and the influence of environmental effects. Fig. 10(c) shows the C-NCMC-SPEG_PBS system is showing a higher *CN* number compared to the other systems, which suggests a higher number of water molecules surrounded by this moiety. This also suggests the stronger interactions of the moiety compared to the other systems. Comparing Fig. 10(a) and (c), the copolymer system shows a higher interaction with PBS compared to the blend system, except for the NCMC_TPEG system. Meanwhile, the B_NCMC_TPEG system shows higher RDF values than the C-NCMC-TPEG system. This suggests the self-assembly of the structure. Comparing Fig. 10(a–d), the first solvation value shows that the PBS system is higher compared to the aqueous media, which is due to the ions present in PBS media.

Fig. 11 shows the RDF between oxygen of the hydroxyl moiety of ciprofloxacin and unreacted alkene group in polymer composites. The first solvation shell and the coordination number are given in Table S4. The first solvation shell is due to electrostatic interactions between the molecules. Comparing Fig. 11(a) and (b) gives a lesser value of solvation

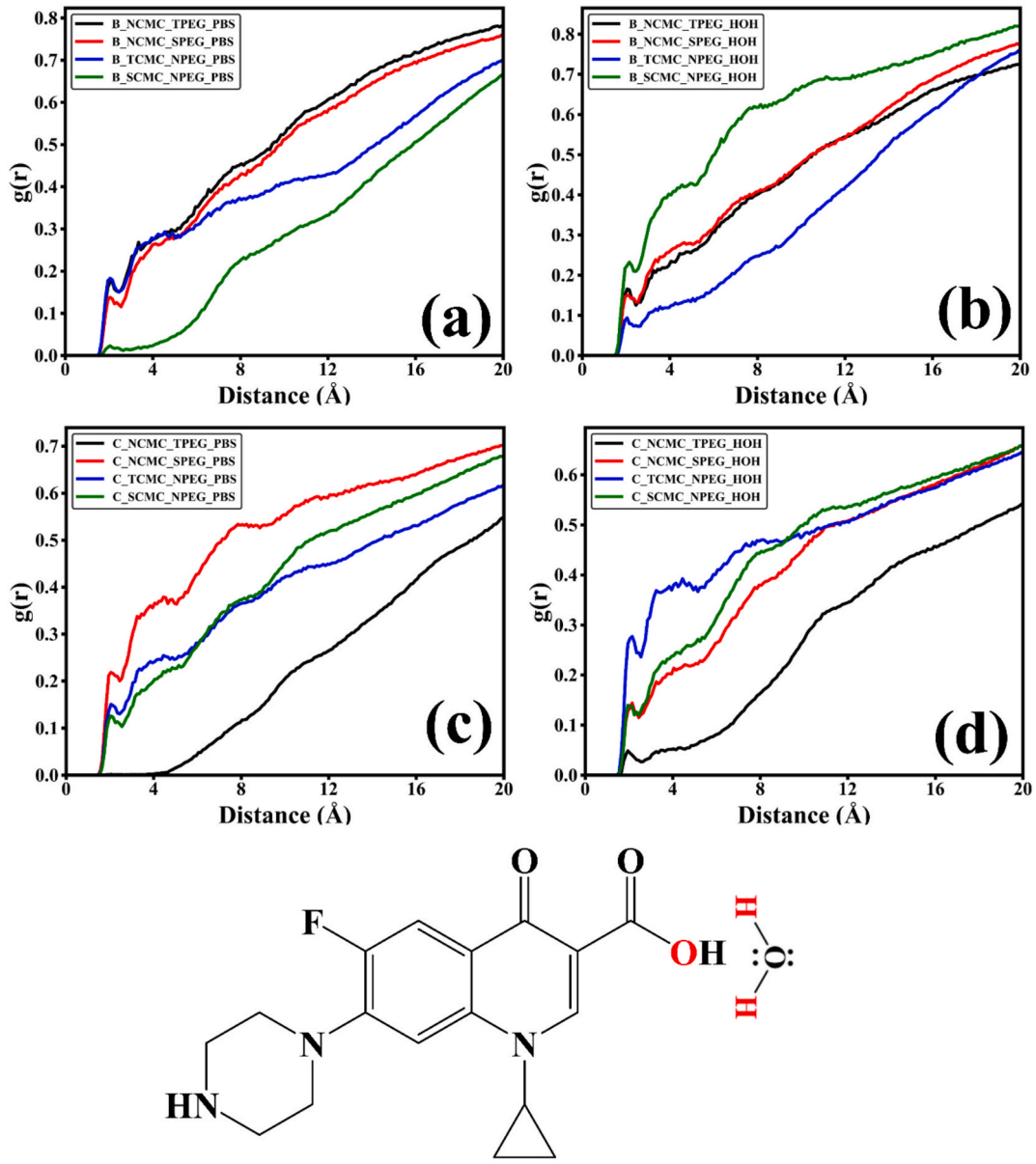


Fig. 10. RDF of oxygen moiety present in the hydroxyl group of the ciprofloxacin to the hydrogen moiety of the water (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous (I) schematic representation of ciprofloxacin and water.

shell for the blend composites with PBS medium as compared to aqueous medium, which implies the role of the PBS ions for the better molecular binding between the drug molecules and the polymer system. This reflects the interaction between the oxygen moiety of the hydroxyl group of the drug and the PBS environment. However, the RDF value of the blend composite system is higher in the aqueous medium. This suggests that the weak electrostatic interactions between the polymer moiety and drug molecules are at longer distances. Comparing Fig. 11(c) and (d), the C-NCMC-TPEG system shows a higher RDF value in the PBS medium compared to the other system present in the PBS and aqueous medium. This is due to the vdW non-bonded interaction between the drug molecules' polymer moieties. Comparing Fig. 11(a–d), the copolymer composites show the highest *CN*, which is related to their higher swelling ratio. The swollen polymer network creates more free space, allowing drug molecules to easily enter and remain near the polymer surface. Although the blend composites form a larger number of H-bonds with the drug, the copolymer mainly holds drug molecules through many weak non-bonded interactions. Even with fewer hydrogen bonds, the higher *CN* values and increased free volume indicate that the copolymer structure can accommodate more drug molecules. Therefore, the copolymer composites demonstrate a higher drug loading capacity than the blend composites. Similarly, Figs. S14–S16 show the solvation shells due to weak electrostatic interactions between molecules. The solvation shell and *CN* are given in Table S4.

Figs. S17 and S18 show the RDFs of the carbonyl moiety of CMC and the ether moiety of the 4-arm PEG, and the hydroxyl moiety of CMC and the ether groups of PEG, respectively. The RDFs for the first solvation shell and *CNs* are given in Table S4. The importance of the first and second solvation shells is a manifestation of van der Waals interactions between the respective moieties. Figs. S17 and S18 present the variation in solvation behavior and *CNs* for the blend and copolymer composites. Elevated *CNs* are a signature of stronger interactions, especially in an aqueous environment, due to the hydrophilic nature of the polymers. Copolymers develop higher RDF values than blends because of the closer proximity of polymer chains, as a result of the copolymerisation process. The RDF between the unreacted moiety (ene) and unreacted moiety (thiol) present in the polymer chain is shown in Fig. S19. The solvation shell and *CNs* are again tabulated using Table S4 In this case, the shell is

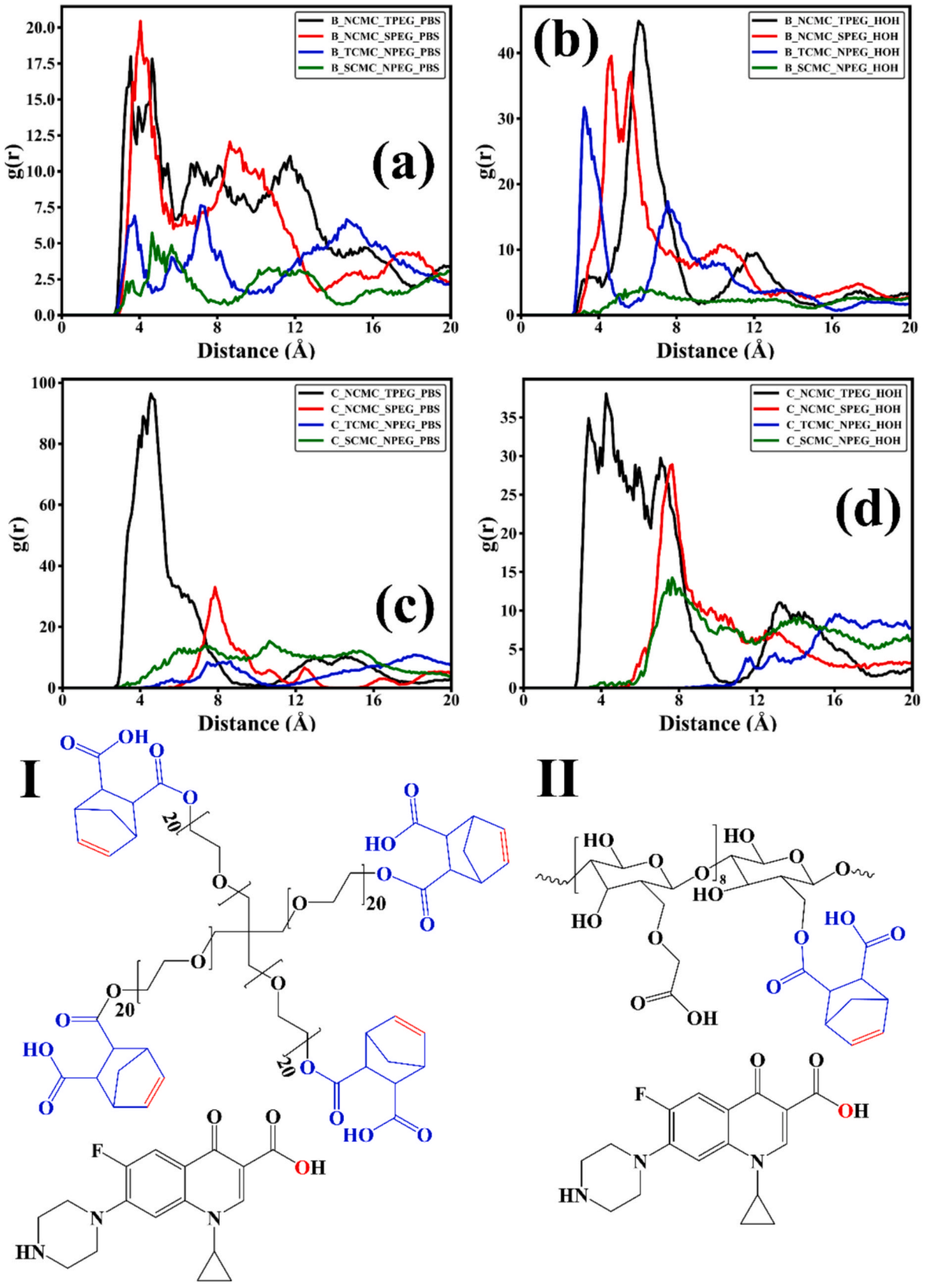


Fig. 11. RDF of oxygen moiety present in the hydroxyl group of the ciprofloxacin to the unreacted ene group present in the polymer framework (a) blend composite in PBS (b) blend composite in aqueous (c) copolymer composite in PBS (d) copolymer composite in aqueous (I) schematic representation of ciprofloxacin and 4-arm-NPEG (II) schematic representation of ciprofloxacin and NCMC.

visible because of van der Waals between molecules. Comparison of Figs. S19(a) and (b) shows that the RDF value was higher in the PBS medium, which suggests the PBS environment is more favourable for the moieties to get closer and have stronger interactions. When comparing Fig. S19(a) and (c), the SCMC_NPEG copolymerisation composites present an increased RDF value in respect to SCMC_NPEG blend composites, which is attributed to the closer arrangement of polymer moieties induced by copolymerisation. Consequently, the value of the RDF of the copolymer composites is greater than that of the blend composites. On the other hand, in the other 3 system, the blend systems have higher RDF values compared with the copolymer systems, which can be explained by polymer self-assembly and dissimilar impacts of the opposite moiety. Accordingly, the features of copolymer composites are that the RDF value is small and the value of the solvation shell is large. A further comparison of Figs. S19(c) and(d) shows that the aqueous medium gives the highest value of RDF, which results from favorable environmental conditions, as well as self-assembly of the polymer.

3.5. Transport of the drug moiety and hydrogel network

Fig. 12 shows the MSD curve of the oxygen atom of the OH functional group of ciprofloxacin in blend composites. The logarithmic MSD slope are given in Table S5. For each system, initially the MSD curve has a linear behaviour with diffusion exponent less than one, indicating sub-diffusive dynamics; when the diffusion exponent is one, it indicates ordinary diffusion. The observed sub diffusion behaviour is due to

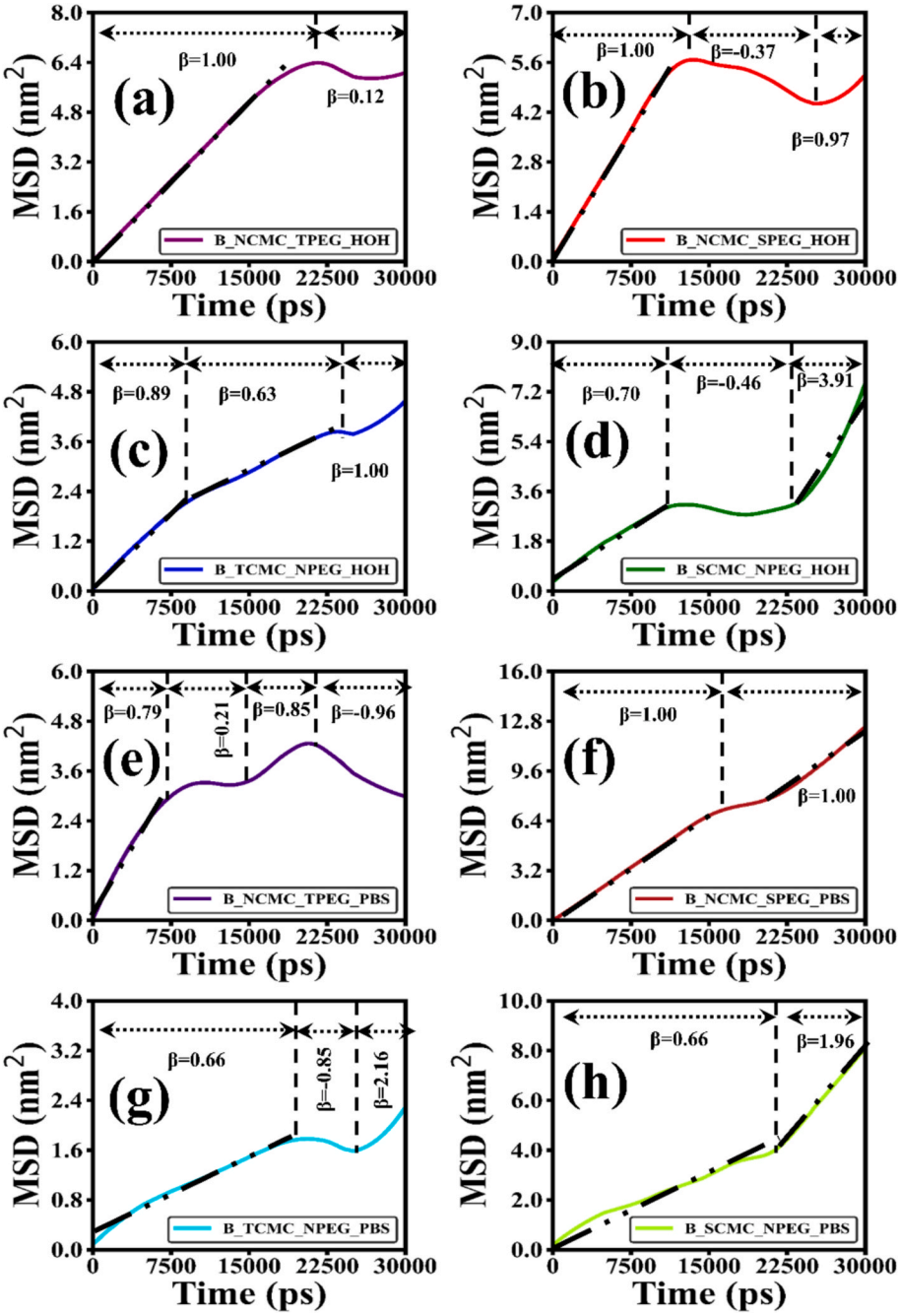


Fig. 12. MSD of oxygen moiety present in the hydroxyl group of the ciprofloxacin in the blend system (a) NCMC_TPEG in the aqueous media (b) NCMC_SPEG in the aqueous media (c) TCMC_NPEG in the aqueous media (d) SCMC_NPEG in the aqueous media (e) NCMC_TPEG in the PBS media (f) NCMC_SPEG in the PBS media (g) TCMC_NPEG in the PBS media (h) SCMC_NPEG in the PBS media.

confinement of drug molecules in the polymer pockets. These pockets form by the structural barrier. Within these local pockets, the particles exhibit restricted diffusion, where the MSD plateaus or grows very slowly with time, reflecting the confinement of the particles. The sub-diffusive behavior is observed for the Fig. 12(c–e) and (g–h). However the other figure shows diffusive behavior. The self-diffusion coefficients for the ciprofloxacin value is 5.70×10^{-11} m^2/s for B_NCMC_TPEG sytem in aqueous media, 8.25×10^{-11} m^2/s for B_NCMC_SPEG system in aqueous media, 7.56×10^{-11} m^2/s for B_NCMC_SPEG sytem in PBS media. The simulated self diffusion coefficient value is lower to compares to the previous reported binary diffusion coefficient in the literature (Dadoo et al., 2017; Stewart, 1996; Wąsik et al., 2013). Self-diffusion describes the random thermal motion of an individual molecule in an equilibrium system, driven only by heat. This motion is slow because it lacks a preferred direction. In contrast, binary diffusion occurs when molecules move due to a concentration or chemical potential gradient, which provides a clear driving force for mass transport. The Darken equation links these two processes by including a thermodynamic factor that accounts for molecular interactions. In well-mixed or stable systems, this factor is greater than one, which enhances transport (Maser, 1999). Therefore, binary diffusion is usually faster than self-diffusion because it combines random motion with an additional thermodynamic driving force. After that time period, Fig. 12(a, e, and g) MSD curve shows non-linear behaviour where the β value for each system is showing nearly zero or less than zero. During this time period, the drug molecules try to escape the structural barrier by finding

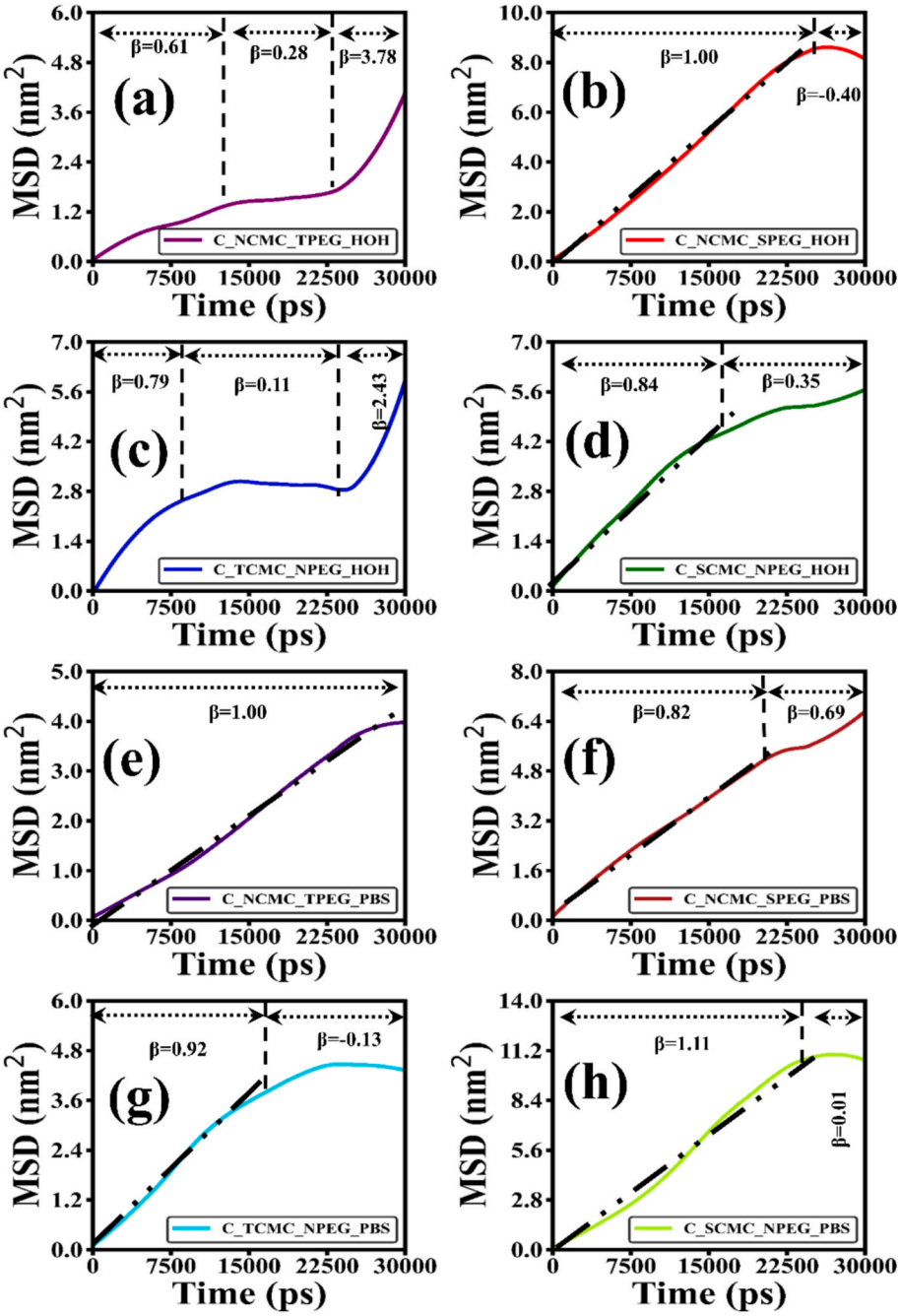


Fig. 13. MSD of oxygen moiety present in the hydroxyl group of the ciprofloxacin in the copolymer system (a) NCMC_TPEG in the aqueous media (b) NCMC_SPEG in the aqueous media (c) TCMC_NPEG in the aqueous media (d) SCMC_NPEG in the aqueous media (e) NCMC_TPEG in the PBS media (f) NCMC_SPEG in the PBS media (g) TCMC_NPEG in the PBS media (h) SCMC_NPEG in the PBS media.

a pathway through the heterogeneous structure. During this time period, the diffusion mechanism is called hindered diffusion. Fig. 12(b, d, and f) shows that the plateau in the MSD curve, which hinders diffusive behaviour. According to Fig. 12(a) and (d–h) the logarithmic slope is greater than one representing the super-diffusion of drug moieties. Such a behaviour could be caused by long range correlations or facilitated transport mechanisms. During these intervals, the slope of the MSD curve is much higher than expected for normal diffusion, suggesting that particles move over larger distances than expected for the time spent doing this for a given time interval. This kind of MSD behaviour is typical for heterogeneous environments. Comparing Fig. 12(a) and (e), the drug molecules, the higher MSD value is observed in the B_NCMC_TPEG system in the aqueous medium compared to the PBS medium. This is due to the higher restricted diffusive behaviour seen in the aqueous medium compared to the PBS medium. Comparing Fig. 12 (b) and (f), a higher MSD value is observed in the PBS medium compared to the aqueous medium. This is observed due to the lower time period of hindered diffusive behaviour in the PBS medium. Comparing the Fig. 12 (c) and (g) the oxygen moiety of the hydroxyl group of the ciprofloxacin shows a higher MSD value in the aqueous medium compared to the PBS medium. The higher MSD value is observed due to the environmental effects. Comparing Fig. 12(d) and (h), the system shows similar MSD values at the end of the simulations in the both media. However, the curve behaviour during dynamics is different. In the aqueous medium, the MSD curve shows restricted diffusion, hindered diffusion and superdiffusive behaviour till the end of the dynamics. In the PBS medium, the hindered diffusion is not observed, which is due to the self-assembly of the drug molecules. Combining the Fig. 12(a)–(h), the highest MSD is observed in the B_NCMC_SPEG system in the PBS medium. This behaviour is observed due to the higher slope value of the restricted diffusive medium and a shorter time period stuck in the hindered diffusive medium. As a result, higher MSD values suggest a lower residence time of the drug molecules.

Fig. 13, the MSD curve represents the oxygen moiety in the carboxylic acid functional group of ciprofloxacin in the copolymer composites. The β value and time periods are detailed in Table S5. Fig. 13(a) and (c) show the initially linear behaviour of MSD, with each system's β value being less than one, which suggests the subdiffusive behaviour. During this period, drug molecules are confined to polymer pockets created by structural barriers. Within these local pockets, the particles display restricted diffusion, leading to a plateau or very slow growth of the MSD with time, indicating particle confinement. During this phase, drug molecules strive to overcome the structural barrier by navigating through the heterogeneous structure, resulting in hindered diffusion. Subsequent to this, the MSD curve of the systems displays linear behaviour, with the β value surpassing one, indicating the superdiffusive behaviour of the molecules. This behaviour can be attributed to long-range correlation or facilitated transport mechanisms. During this time, the MSD curve depicts a steeper slope compared to normal diffusion, indicating that the particles are travelling longer distances than expected for a given time interval. This type of MSD behaviour is observed in heterogeneous environments. Fig. 13(b) and(e) demonstrate the linear behaviour with a β value is equal to one, which suggests diffusive behaviour. The self-diffusive coefficient value is 5.67×10^{-11} m^2/s for C_NCMC_SPEG sytem in aqueous media, 2.29×10^{-11} m^2/s for B_NCMC_TPEG sytem in PBS media.. Fig. 13(d) demonstrates the linear behaviour of MSD initially, with the system's β value being less than one, which suggests the subdiffusive behaviour. This behaviour is observed due to the drug molecules being stuck in the polymer pockets. After that, the moiety shows a negative slope of MSD, which suggests the hindered diffusive behaviour where indicating the particle's confinement. In Fig. 13(e), (f), and (g), it is evident that the system's β value is less than one, indicating subdiffusive behaviour, which is attributed to the drug molecules being trapped in the polymer pockets. However, after a certain time period, the MSD curve changes its slope, but the β value stays less than one, which suggests the subdiffusive behaviour. In comparing Fig. 13(a–e), show a higher MSD value is observed in the C-NCMC-SPEG system in the aqueous medium compared to the other systems. This higher MSD value is observed due to the superdiffusive behaviour. In comparing Fig. 13(f–h), a higher MSD value is observed in the C-SCMC-NPEG system in the aqueous medium compared to the other systems, which are observed due to the superdiffusive behaviour. The higher MSD value is observed in the PBS medium compared to the aqueous medium. This is due to the higher restricted diffusive slope values in the system. This may indicate the lower residence time of the drug molecules.

Figs. S20 and 21 show the MSD of the unreacted ene group present in the norbornene structure and the unreacted thiol group present in the polymer blend composites, respectively. The β value and time periods are given in Table S5. Looking at the β values in the Figure, shows restricted diffusive behaviour, hindered diffusive behaviour and superdiffusive behaviour. Comparing Fig. S20(a–h), a higher MSD is observed in the B_NCMC_SPEG system in the PBS medium. This is due to the self-assembly of the polymer. It also suggests that the higher MSD results in a higher rate of diffusion of the polymer moiety compared to the other system, which suggests a lower residence time. However, higher MSD is observed initially in the B_NCMC_SPEG system in the aqueous medium. During this time period, the unreacted ene moiety shows restricted diffusive behaviour with a high slope value of 9.2 ns. During this time period, the molecules are stuck in the structure where every molecule tries to interact with the system. After that, it shows hindered diffusive behaviour till the end of the dynamics, which is due to trying to break the structural barrier. Figs. S22 and S23 depict the MSD of the unreacted ene and thiol groups present in the copolymer composites, respectively, showing restricted diffusive, hindered diffusive, and superdiffusive behaviours. Analysis within the Figs. S22 and S23 series indicates a higher MSD in the C-SCMC-NPEG system in the PBS medium, attributed to the self-assembly of the polymer and suggesting a higher rate of diffusion of polymer groups compared to the other systems, indicating a lower residence time. The drug molecules and polymer matrix MSD curves are quite similar due to the non-bonded interactions between the drug molecules and the polymer matrix. However, the drug molecules demonstrate higher mobility compared to the polymer matrix due to the greater number of H-bonds with solvent molecules.

The MSD analysis provides critical insights into drug–polymer dynamics, guiding hydrogel design for optimized drug delivery performance. The observed subdiffusive, hindered, and superdiffusive behaviors reveal how polymer network architecture controls molecular mobility and release kinetics. Subdiffusive behavior, driven by drug confinement within polymer pockets, indicates strong drug–matrix interactions suitable for sustained or controlled release applications. Conversely, superdiffusive behavior, characterized by enhanced molecular transport, suggests pathways for rapid or burst release formulations. Systems such as B_NCMC_SPEG and C_NCMC_SPEG showing higher MSD values and lower residence times highlight how functional group composition and solvent environment dictate diffusion rates. For material engineers, tailoring crosslinking density, hydrophilicity, and functional moieties (e.g., thiol, carboxylic, or hydroxyl groups) allows precise modulation of diffusion barriers and release duration. Pharmaceutical formulators can leverage these findings to design hydrogels with tunable release profiles—achieving either rapid delivery for acute treatments or prolonged, steady diffusion for chronic therapies. Moreover, understanding solvent-dependent diffusion behaviors (PBS vs. aqueous media) enables rational selection of in vivo environments and buffer systems for predictable release kinetics. Ultimately, integrating MSD-based molecular transport analysis into hydrogel design ensures more effective, stable, and patient-specific drug delivery systems with optimized therapeutic performance.

4. Conclusion

All-atom molecular dynamics simulations combined with molecular

docking provided critical insights into the structural and dynamic behavior of ciprofloxacin-loaded thiol-ene 'click' CMC and 4-arm-PEG hydrogels. The study reveals a fundamental design trade-off between the two architectures: while blend composites exhibit superior structural stability and chain entanglement—indicated by lower RMSD and MSD values that favor prolonged drug residence times—the copolymer systems offer enhanced swelling ratios and stronger drug-polymer interactions. Radial distribution function (RDF) analysis confirms that the copolymer architecture facilitates robust solvation shells around ether moieties, promoting higher drug-loading capacities and tighter molecular confinement. Notably, the release kinetics are highly responsive to environmental factors, with the transition from aqueous to PBS media significantly altering diffusion pathways due to ionic interactions and the kosmotropic nature of PEG in the blend systems. Although these findings highlight the tunability of thiol-ene hydrogels for controlled delivery, the observed mechanisms are intrinsically linked to ciprofloxacin's specific physicochemical properties. Therefore, these results should be viewed as a baseline for future investigations rather than a universal rule. To fully capitalize on this platform, future experimental and computational studies must expand this framework to therapeutic agents with diverse polarities and molecular sizes. Ultimately, this research demonstrates that by judiciously selecting between copolymer and blend strategies, engineers can precisely tailor hydrogel properties—balancing high loading capacity against sustained release profiles-to meet the rigorous demands of advanced biomedical applications.

Ethical approval

Not applicable.

CRediT authorship contribution statement

Rontu Das: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Debashis Kundu:** Writing – review & editing, Visualization, Supervision, Software, Resources, Project administration, Funding acquisition, Conceptualization.

Informed consent

Not applicable.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Debashis Kundu reports financial support was provided by Science and Engineering Research Board in India. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ces.2026.123356.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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